

Dwight Look College of ENGINEERING
TEXAS A&M UNIVERSITY

ECEN 403
Status Update Presentation

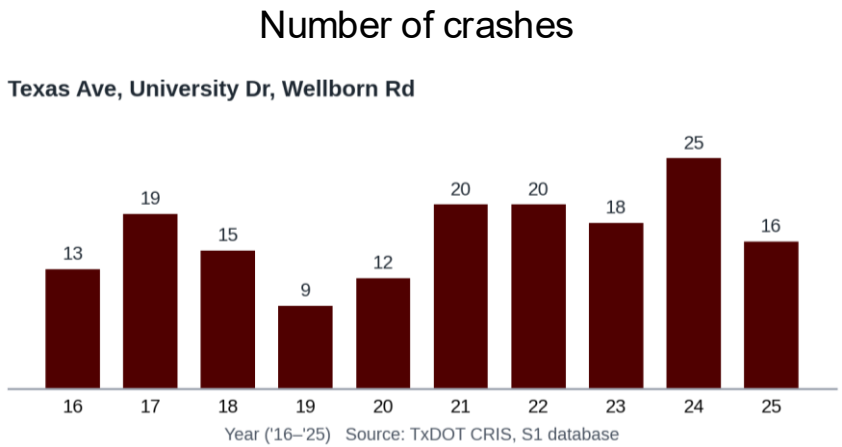
Team 4: AI Traffic-Safety Consultant

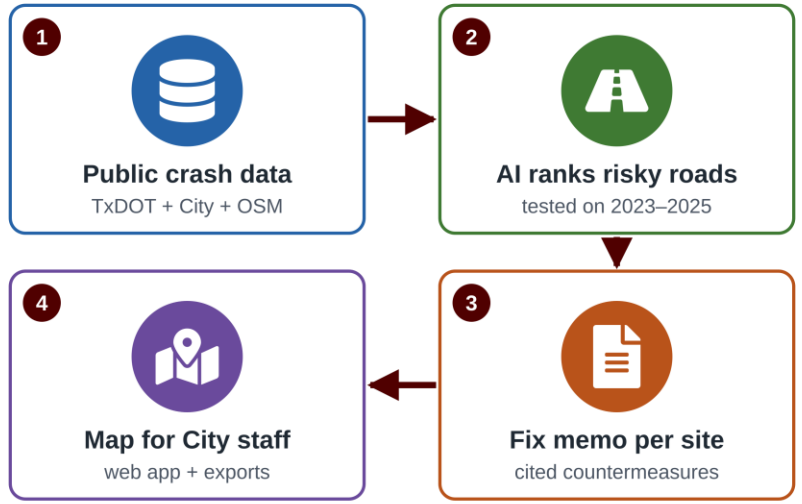
Param Shelar, Kabir Singh, Jose Ruiz Garcia, Juan Lopez

Problem and Scope

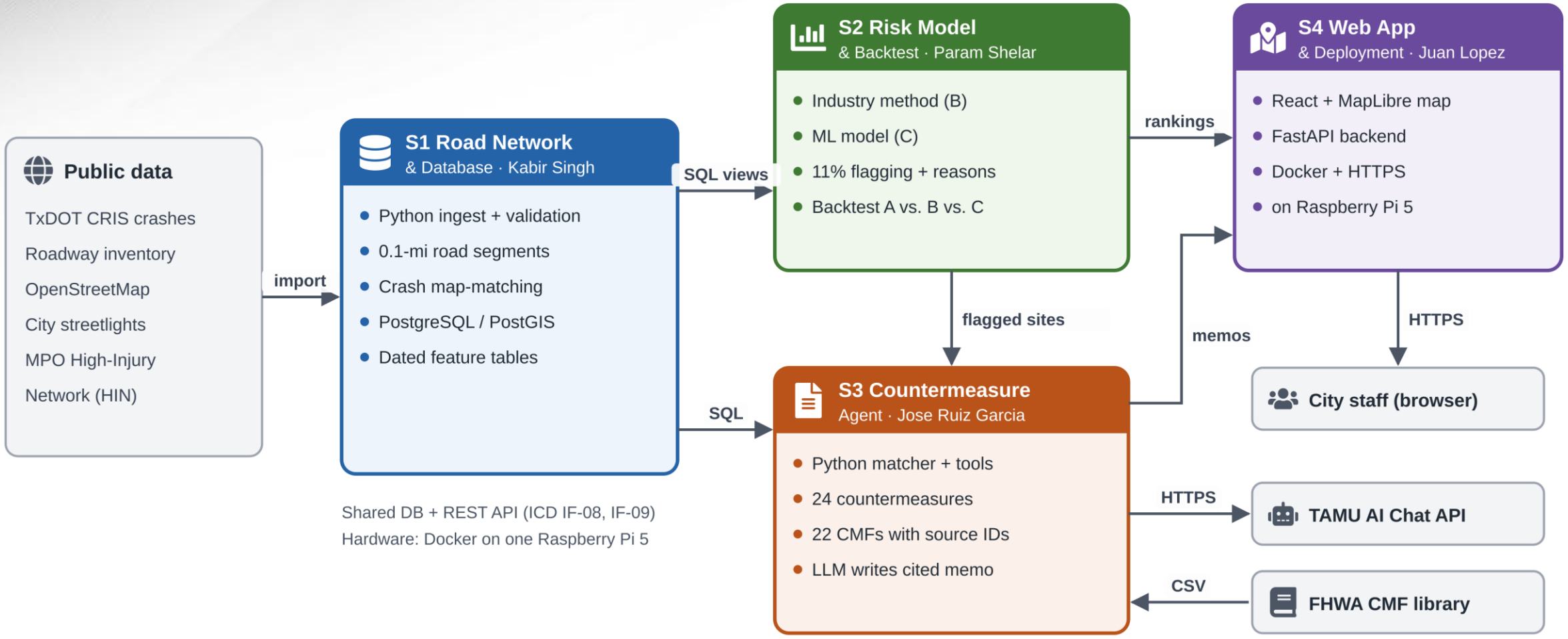
Problem statement:

- 625+ severe crashes in Brazos County: 100+ fatal, the rest serious injuries
- Goal: identify College Station's highest-risk roads and give the city a plan to reduce crashes
- System capabilities:
  - –Rank every road by crash risk
  - –Back test rankings against historical crashes
  - –Generate a safety plan for the city
  - –Website to display findings





System Overview



Subsystem 1: Road Network and Database

Kabir Singh

Accomplishments since midterm update 30 hrs of effort	Ongoing progress/problems and plans until the next presentation
• PostGIS database running; its guard blocks post-2022 data (14/14 tests pass) • 16,539 College Station crashes loaded (2016–2025), 0% bad records • 3 corridors split into 185 road segments, 18.1 miles • 98.1% of severe crashes matched to a road (target: 95%)	• Read-only views so the risk model can run (M2, Oct 23) • Load the consultant's HIN map layer • Grow from 3 corridors to the full city network • Add OpenStreetMap and streetlight features

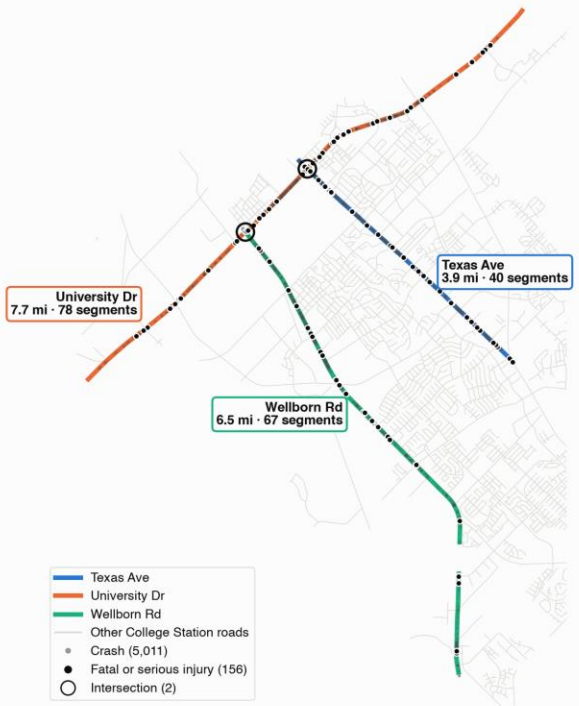
Final demo: full-city database with HIN, OpenStreetMap and streetlight data, all tests passing

Subsystem 1: Road Network and Database

Kabir Singh

Three-corridor road network with matched crashes

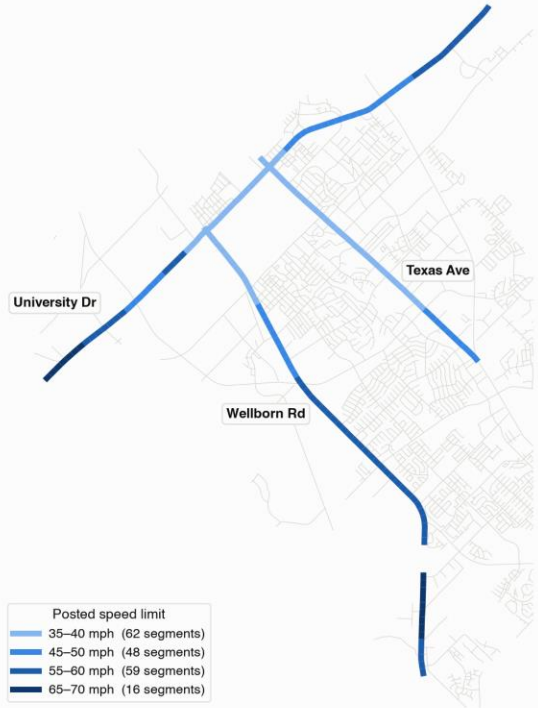
18.1 miles cut into 185 segments of about 0.1 mi; 5,167 crashes from 2016–2025 placed on the network.



Source: TxDOT CRIS crash records 2016–2025 (City of College Station); TxDOT Roadway Inventory 2019. TASC S1 database, network v0_2019.

Feature table: posted speed limit on each segment

Speed limit is one of 5 features stored for every segment (925 values), all dated 2019-12-31 and checked against the 2022 leakage cutoff.



Source: TxDOT CRIS crash records 2016–2025 (City of College Station); TxDOT Roadway Inventory 2019. TASC S1 database, network v0_2019.

Subsystem 2: Risk Model and Back test

Param Shelar

Accomplishments since midterm update 35 hours of effort	Ongoing progress/problems and plans until the next presentation
• HIN: number of serious crashes • Industry standard practice; road condition, speed limit etc. • Our machine learning model • 27 core files: 17 python modules, 10 test modules • All code works, tested	• Fine tune models; currently performing poorly • Explanations using SHAP; crash occurred because of feature x

File	What it does
nb_spf.py	Method B safety model
ml_risk.py	Method C ML model
hin_baseline.py	Method A HIN lookup
empirical_bayes.py	Crash-history blending (EB)
severity.py	Serious-crash probability
ranking.py	Ranks and flags the top 11%
backtest.py	Grading with bootstrap intervals
explanations.py	Per-site reasons (SHAP)
preprocessing.py	Feature cleaning and encoding
leakage.py	Bad-input and future-data blocking
bit.py	Built-in self-checks
train.py	Full training pipeline
cli.py	Command-line entry point
artifacts.py	Saving runs, checksums, replay
database.py	Reads S1's database
export_payload.py	Sends results back to S1
make_demo.py	Practice-data generator

Subsystem 2: Risk Model and Back test

Param Shelar

```
{'event': 'bit_pass'}
{'event': 'snapshot_validated', 'sites': 1000, 'features': 5000, 'history_rows': 7000, 'test_crashes': 1851, 'hin_available': True}
{'event': 'features_prepared', 'sites': 1000, 'panel_rows': 7000, 'encoded_features': 14, 'duration_seconds': 0.47933725899997626}
{'event': 'nb_spf_fitted', 'k': 0.494592687287664, 'duration_seconds': 0.026715906999982053}
{'event': 'ml_eb_fitted', 'k': 0.5413814674603382, 'k_method': '5-fold out-of-fold site predictions', 'duration_seconds': 0.12568829200000664}
{'event': 'severity_fitted', 'calibration_status': 'ok', 'duration_seconds': 0.00719025800003692}
{'event': 'predictions_and_shap_checked', 'explanations': 550, 'duration_seconds': 0.06479250399996772}
{'event': 'backtest_scored', 'resamples': 1000, 'n_test_fsi': 103, 'duration_seconds': 0.17584892000002128}
{'event': 'run_complete', 'path': 'outputs/practice-1000', 'duration_seconds': 1.2935867570000141}
backtest.json  calibration.json  complete.json  diagnostics.json  explanations.json  manifest.json  ml_parameters.json  nb_spf.json       predictions_B.csv  severity.json
bit_log.json   calibration.svg   config.json    encoder.json      features.json      ml_model.ubj   model_card.md       predictions.json  predictions_C.csv  snapshot.json
(.venv) paramshelar@paramshelar:~/Desktop/AICorps4/Param_Shelar$
```

```
 rank site_id  fsi_rate  flagged
  108   s0305  0.664104     True
  109   s0800  0.663536     True
  110   s0872  0.659771     True
  111   s0615  0.655756    False
  112   s0018  0.653099    False
  113   s0936  0.652863    False
flagged
False    89.0
True     11.0
Name: network_length_mi, dtype: float64
increases | 24475 vehicles/day daily traffic
increases | 55 mph posted speed
increases | 5 TxDOT functional class
decreases | 2 traffic lanes
neutral | undivided median type
>>>
```

```
TOTAL                                        1674    310    81%
84 passed, 20 subtests passed in 1.29s
{"event": "replay_pass", "run_id": "practice-1000", "duration_seconds": 0.9341455410000208}
```

site_id	rank	speed_mph	lanes	daily_traffic	serious_2016_22	predicted_2023_25	actual_2023_25
s0613	1	55	2	24475	2	0.79	1
s0663	2	55	6	22905	2	0.76	1
s0211	3	55	4	23878	3	0.75	1
s0956	4	55	6	23923	2	0.74	1
s0493	5	55	6	21155	2	0.64	0
s0813	6	55	6	23096	2	0.57	2
s0859	7	55	6	22923	3	0.57	1
s0684	8	55	2	15834	0	0.50	0
s0132	9	45	2	18824	0	0.50	1
s0297	10	55	2	6983	1	0.48	1

```
Top 10 sites: 0.90 hidden serious crashes per site
All 1000 sites: 0.10 per site
```

Subsystem 3: Countermeasure Agent

Jose Ruiz Garcia

Accomplishments since midterm update 33 hrs of effort	Ongoing progress/problems and plans until the next presentation
•Built a library of 24 countermeasure and planning entries. •Recorded 22 crash modification factors (CMFs) with source IDs and study conditions. •Developed a Python matcher that suggests candidate treatments and explains why they match. •Passed all 12 automated tests covering matching rules and safeguards.	•Finalize the input/output format with S1 and S2. •Complete CMF source and applicability verification. •Test the matcher on a real site flagged by S2. •Develop a draft AI memo with traceable sources and checked numerical claims.

Final demo goal: matcher running on real flagged sites; first AI memo passing the grounding check

Subsystem 3: Countermeasure Agent

Jose Ruiz Garcia

Countermeasure library

cm_id	psc_name	cmf_id	cmf	crash_type	severity_sc	site_type	star_rating
CM001	Crosswalk	4123	0.6	pedestrian	All	signalized i	2
CM002	Leading Pe	9918	0.87	pedestrian	All	signalized i	5
CM003	Medians ar	8799	0.685	pedestrian	All	segment;u	4
CM004	Pedestrian	9020	0.453	pedestrian	All	segment;u	3
CM005	Rectangula	9024	0.526	pedestrian	All	segment;u	3

Example Input

```
{
  "site_id": "SYN0012_SIDEWALK",
  "site_type": "segment",
  "crash_patterns": [
    "pedestrian"
  ],
  "area_type": "urban",
  "severity_scope": "All"
}
```

Matcher Output

```
  CM006: Install sidewalk to avoid walking along roadway
    MATCH: Location type matches: segment.
    MATCH: CMF-supported crash pattern matches: pedestrian.
    REVIEW: Area setting needs review: study=Not specified.
    REVIEW: Check treatment prerequisites, geometry, traffic volumes, and existing equipment: Targets pedestrians
    REVIEW: Complete source verification; current status: evidence_checked_unrated
    REVIEW: Research quality is officially unrated.
  CM023: Convert four-lane undivided road to two through lanes plus center turn lane
    MATCH: Location type matches: segment.
    MATCH: Area setting is compatible with the recorded study setting.
    MATCH: Study covers all crash types combined.
    REVIEW: Aggregate CMF does not establish a reduction for the requested crash subtype alone.
    REVIEW: Check treatment prerequisites, geometry, traffic volumes, and existing equipment: Existing four-lane
    REVIEW: Complete source verification; current status: evidence_checked_indexed
Candidates: 6
```

12/12 test passed

```
✔ Tests passed: 12 of 12 tests – 7 ms

Ran 12 tests in 0.028s

OK

Process finished with exit code 0
```

Subsystem 4: Web App and Deployment

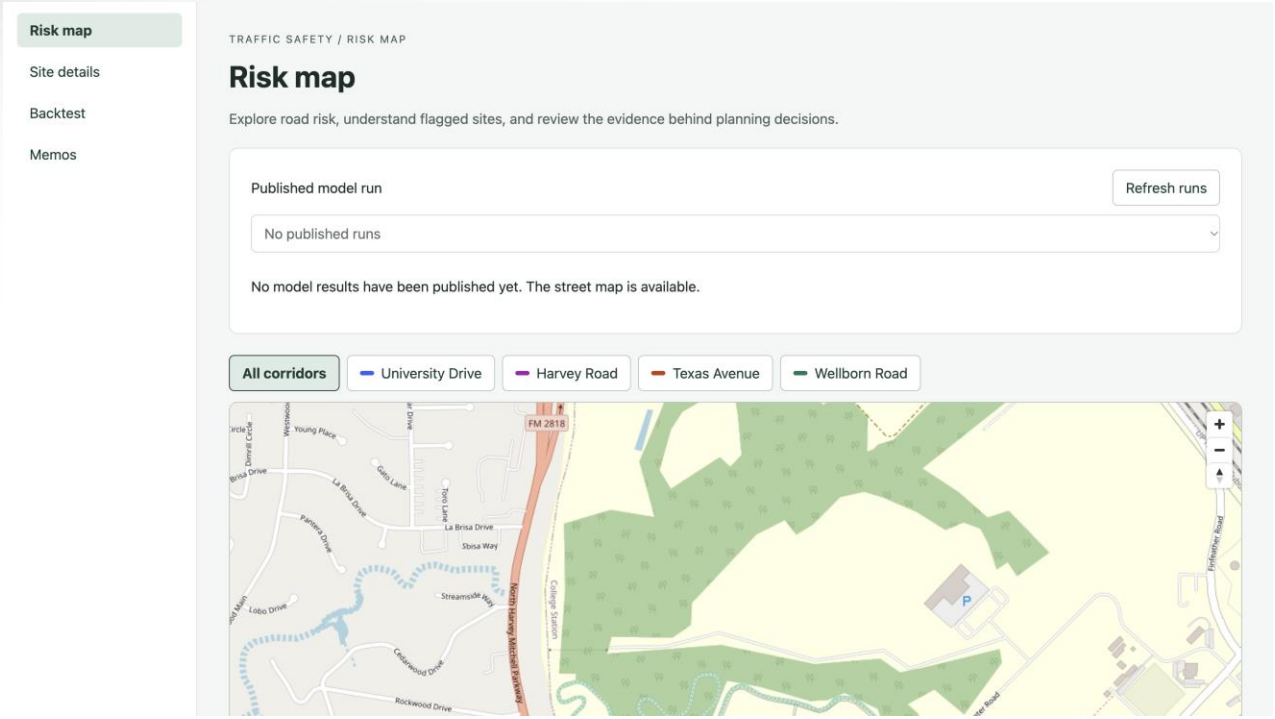
Juan Lopez

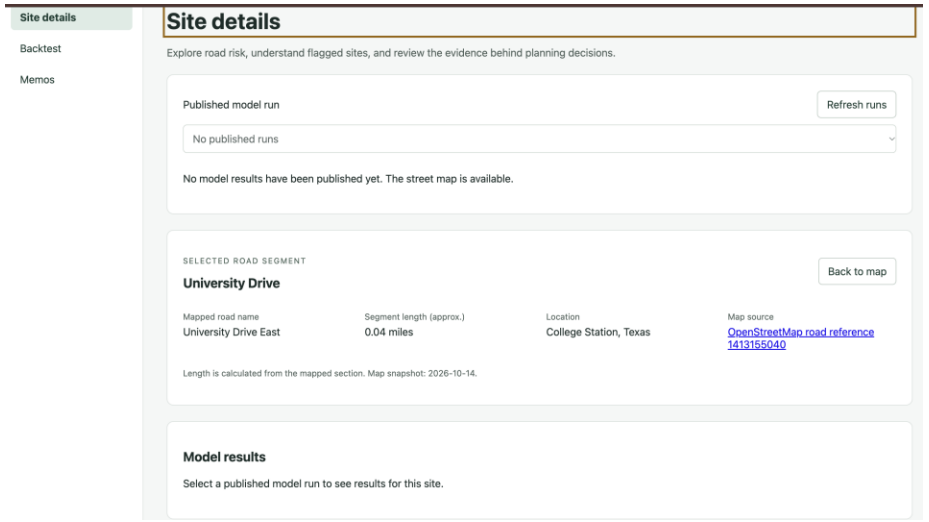
Accomplishments since midterm update 29 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Interactive map of four corridors with focus controls and clickable road details • Backtest and Memos tabs enabled • FastAPI JSON adapter validates and loads model runs, rankings, site results, backtests and memos • Loading, retry and missingresult states • 8 backend tests, frontend build and sample data browser checks pass	• Connect the shared database and real S2 model outputs • Match stable site IDs and draw traffic risk layers • Connect S3 memo generation and file exports • Add authentication, containers and HTTPS on the Pi 5 • Complete performance, accessibility and full integration tests

Final demo: map and site pages showing real ranked sites; backtest page live

Subsystem 4: Web App and Deployment

Juan Lopez

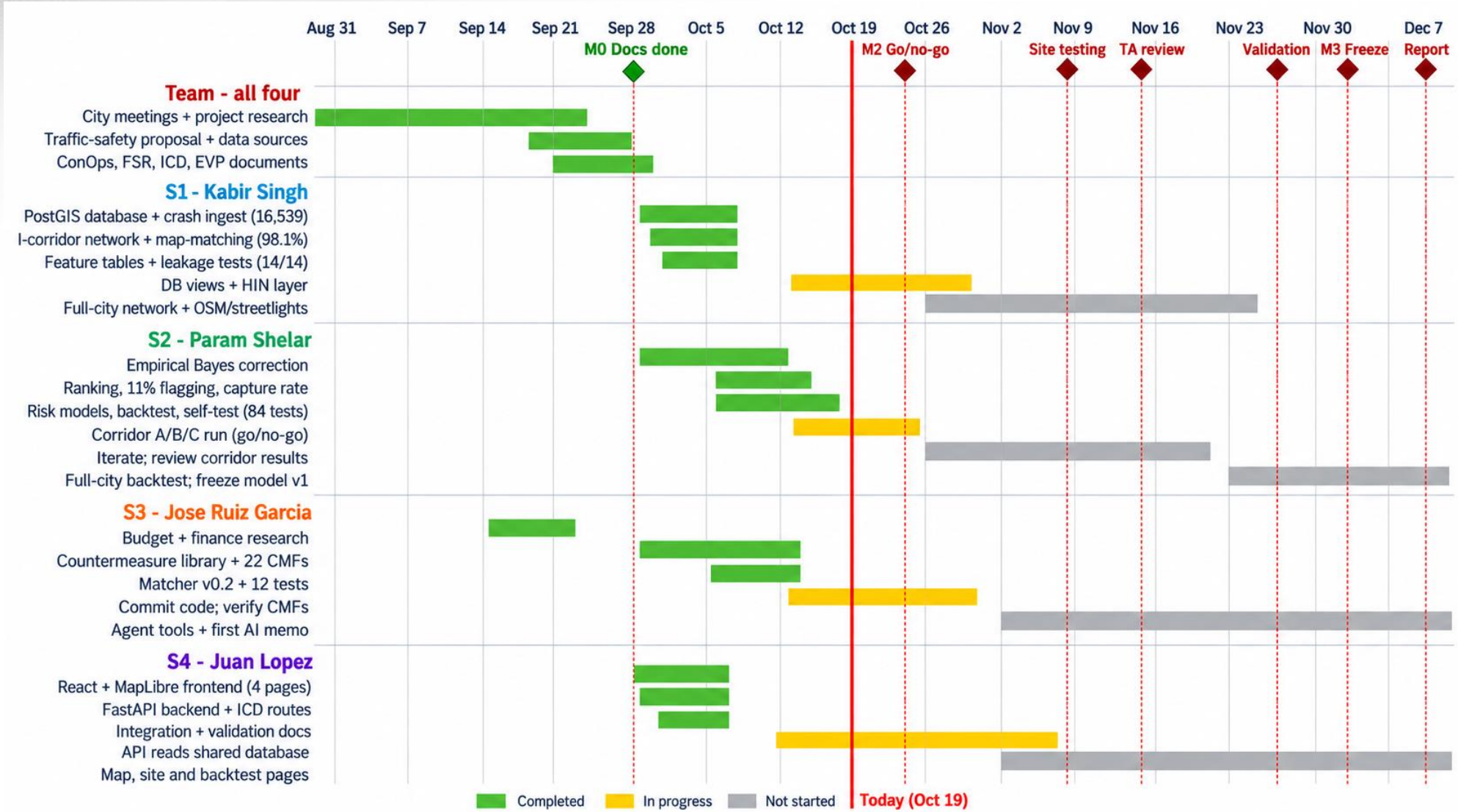




Backtest and Memo tabs active, Basic layout for AI integration model installed

Risk map with University Drive, Harvey Rd, Texas Avenue and Wellborn Rd

Execution


Aug 31
Sep 7
Sep 14
Sep 21
Sep 28
Oct 5
Oct 12
Oct 19
Oct 26
Nov 2
Nov 9
Nov 16
Nov 23
Nov 30
Dec 7
M0 Docs done
M2 Go/no-go
Site testing
TA review
Validation
M3 Freeze
Report
Team - all four
City meetings + project research
Traffic-safety proposal + data sources
ConOps, FSR, ICD, EVP documents
S1 - Kabir Singh
PostGIS database + crash ingest (16,539)
I-corridor network + map-matching (98.1%)
Feature tables + leakage tests (14/14)
DB views + HIN layer
Full-city network + OSM/streetlights
S2 - Param Shelar
Empirical Bayes correction
Ranking, 11% flagging, capture rate
Risk models, backtest, self-test (84 tests)
Corridor A/B/C run (go/no-go)
Iterate; review corridor results
Full-city backtest; freeze model v1
S3 - Jose Ruiz Garcia
Budget + finance research
Countermeasure library + 22 CMFs
Matcher v0.2 + 12 tests
Commit code; verify CMFs
Agent tools + first AI memo
S4 - Juan Lopez
React + MapLibre frontend (4 pages)
FastAPI backend + ICD routes
Integration + validation docs
API reads shared database
Map, site and backtest pages
Completed
In progress
Not started
Today (Oct 19)

Validation plan

Test name	Success criteria	Methodology	Status	Responsible engineer
Map-matching rate (FSR 3.2.1.4)	≥ 95% of severe crashes matched	Test	Met: 98.1% (corridors)	Kabir Singh
Temporal leakage (FSR 3.2.1.5)	Post-2022 feature blocks the run	Test	Passing (14/14 tests)	Kabir Singh
High-risk flagging (FSR 3.2.1.9)	Flagged length = 11% ± 0.5%	Test	Built; limited data only	Param Shelar
Backtest reporting (FSR 3.2.1.10)	A/B/C rates with 1,000-resample CIs	Analysis	Built; real run at M2	Param Shelar
Reproducibility (FSR 3.2.1.21)	Re-run gives identical ranking	Test	Passing (practice run)	Param Shelar
Numeric grounding (FSR 3.2.1.14)	20/20 planted fake numbers blocked	Test	Not started (M5)	Jose Ruiz Garcia
Citation grounding (FSR 3.2.1.15)	Every fix cites a CMF or FHWA ID	Test	In progress: 24 CMF IDs	Jose Ruiz Garcia
Web app functions (FSR 3.2.1.18)	Map, site, backtest pages work	Demo	Shells built (M6)	Juan Lopez
Web responsiveness (FSR 3.2.1.19)	Map ≤ 5 s, site ≤ 2 s, 5 users	Test	Not started (M7)	Juan Lopez

Status: green = completed / passing · yellow = in progress · gray = not started

Thank You

Team 4: AI Traffic-Safety Consultant

That’s everything. Thank you for your time; we are open to questions.